

Lesson 1 · Gradients, Differentiation and Tangents

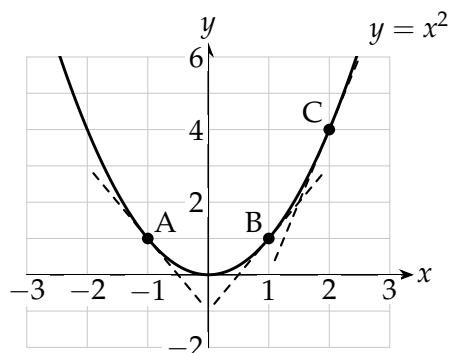
NCEA Level 2 Mathematics · AS 91262 Apply calculus methods in solving problems · Handout

1 The one idea

A straight line has one gradient. A curve does not: its steepness changes from point to point. The **gradient of a curve at a point** is the gradient of the tangent at that point (the straight line that just touches the curve there).

$f'(x)$ is a **gradient machine**: put in an x -value, get out the gradient of the curve at that point.

For $y = x^2$ the gradient machine is $f'(x) = 2x$. At A ($x = -1$) it gives -2 (going down), at B ($x = 1$) it gives 2 , and at C ($x = 2$) it gives 4 (steeper than at B).



2 Words and symbols

gradient function

the function that gives the gradient at every x . Also called the **derivative**.

$f'(x)$ or $\frac{dy}{dx}$

two ways to write the gradient function. Say “ f dash of x ” and “ dy by dx ”.

differentiate

find the gradient function.

tangent

the straight line touching the curve at one point, with the same gradient as the curve there.

3 Differentiating: the power rule

$$\frac{d}{dx}(ax^n) = an x^{n-1} \quad \text{multiply by the power, then take 1 off the power.}$$

Differentiate each term on its own.

term	what happens	derivative	why
x^5	power 5 comes to the front, power becomes 4	$5x^4$	rule
$4x^3$	$4 \times 3 = 12$, power becomes 2	$12x^2$	rule
$7x$	$x = x^1$, so $7 \times 1 \times x^0$	7	an x -term leaves its number
9	a constant does not change	0	a flat line has gradient 0

Expand brackets first. The power rule works on single terms only.

Worked example 1

Differentiate $f(x) = 3x^4 - 5x^2 + 7x - 2$.

Step 1 Term by term: $3x^4 \rightarrow 12x^3$, $-5x^2 \rightarrow -10x$, $7x \rightarrow 7$, $-2 \rightarrow 0$.

Step 2 $f'(x) = 12x^3 - 10x + 7$

Worked example 2

Differentiate $y = (2x + 1)(x - 3)$.

Step 1 Expand: $y = 2x^2 - 6x + x - 3 = 2x^2 - 5x - 3$.

Step 2 Differentiate: $\frac{dy}{dx} = 4x - 5$.

Check: multiplying the derivatives of the brackets ($2 \times 1 = 2$) is wrong, see L1 below.

4 Three question types

Read what is *given* and what is *asked*; that decides the route.

given $\rightarrow$ asked	route	watch
$x \rightarrow$ gradient	differentiate, substitute x into $f'(x)$	not into $f(x)$
gradient $\rightarrow$ point	solve $f'(x) = \text{gradient}$ for x , then y from $f(x)$	a point needs x and y
gradient at $x \rightarrow$ unknown constant	differentiate with the letter in, substitute x , set equal to the gradient, solve	the letter stays in $f'(x)$

Worked example 3 · gradient at a point

Find the gradient of $f(x) = x^3 - 6x^2 + 4$ at the point where $x = 5$.

Step 1 $f'(x) = 3x^2 - 12x$

Step 2 $f'(5) = 3(5)^2 - 12(5) = 75 - 60 = 15$

Worked example 4 · point with a given gradient

Find the point on $y = x^2 - 7x + 1$ where the gradient is 3.

Step 1 $\frac{dy}{dx} = 2x - 7$

Step 2 $2x - 7 = 3$, so $x = 5$.

Step 3 $y = 5^2 - 7(5) + 1 = -9$. The point is $(5, -9)$.

Worked example 5 · unknown constant (Merit step)

The curve $f(x) = ax^2 + 3x - 1$ has gradient 11 at $x = 2$. Find a .

Step 1 $f'(x) = 2ax + 3$ (treat a like a number)

Step 2 $f'(2) = 4a + 3 = 11$

Step 3 $4a = 8$, so $a = 2$. Check: $f'(2) = 2(2)(2) + 3 = 11$.

5 Equation of a tangent

Write it in this order

1. **Point:** x_1 is given; find y_1 from $f(x)$.
2. **Gradient:** $m = f'(x_1)$, a number.
3. **Line:** $y - y_1 = m(x - x_1)$, then rearrange to $y = mx + c$.

Worked example 6

Find the equation of the tangent to $y = x^3 - 2x + 1$ at $x = 2$.

Step 1 Point: $y = 2^3 - 2(2) + 1 = 5$, so $(2, 5)$.

Step 2 Gradient: $\frac{dy}{dx} = 3x^2 - 2$, so $m = 3(2)^2 - 2 = 10$.

Step 3 Line: $y - 5 = 10(x - 2)$, so $y = 10x - 15$.

Worked example 7 · two unknowns (Merit)

The curve $y = ax^2 + bx$ passes through $(1, 5)$, and its gradient at that point is 8. Find a and b .

Step 1 On the curve: $a(1)^2 + b(1) = 5$, so $a + b = 5$.

Step 2 Gradient: $\frac{dy}{dx} = 2ax + b$; at $x = 1$: $2a + b = 8$.

Step 3 Subtract: $a = 3$, then $b = 2$. $a = 3$, $b = 2$.

Two unknowns need two equations: one from the **point**, one from the **gradient**.

Parallel lines have equal gradients. “The tangent is parallel to $y = 2x + 1$ ” means $f'(x) = 2$.

6 What each grade looks like here

Achieved	apply one method: differentiate and find a gradient, or find a tangent at a given x .
Merit	link steps (relational thinking): find an unknown constant from a gradient, use two conditions for two unknowns, use a parallel line.
Excellence	extended abstract thinking: e.g. a tangent through an outside point, or where a tangent meets the curve again, written as a full chain of reasoning.

7 Checking with the fx-9750GIII

- **Gradient at a point:** RUN-MATRIX $\rightarrow$ OPTN $\rightarrow$ CALC (F4) $\rightarrow$ d/dx (F2). Type the function in X, a comma, then the x -value: e.g. $d/dx(X^3 - 2X + 1, 2)$ gives 10.
- **Solving a quadratic:** EQUA $\rightarrow$ POLY (F2) $\rightarrow$ degree 2, enter the coefficients.
- The calculator checks the answer. The written $f'(x)$ is what shows the method.

8 Ways to lose marks

From the material plus teaching observation.

- L1** Differentiating brackets without expanding first: $(2x + 1)(x - 3)$ is not differentiated factor by factor.
- L2** Using the wrong function: the **gradient** comes from $f'(x)$, the **y -coordinate** comes from $f(x)$.
- L3** Slips on simple terms: $7x \rightarrow 7$ (not $7x$ or 0), and a constant $\rightarrow 0$.
- L4** Leaving x inside the tangent's gradient: $y - y_1 = (4x - 3)(x - x_1)$ is not a straight line. Substitute first.
- L5** Sign slips with negative x : write $(-1)^2$ and $(-1)^3$ in brackets.
- L6** “Use calculus” ignored: $f'(x)$ must be written down. A calculator value alone does not show the method.